Trip-Length Distributions Enable Recovery of Distance-Decay Parameters for Zero-Shot Gravity Calibration

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Abstract

Predicting commuting flows without individual travel trajectories is a major challenge in urban planning. While deep learning models achieve high accuracy, they require localized origin-destination (OD) surveys for calibration, limiting deployment in data-scarce or privacy-restricted regions. This study shows that aggregate trip-length distributions (TLDs)—population-level travel distance histograms—are sufficient to recover city-specific distance-decay parameters without exposing individual trajectories. We evaluate an OD-free calibration setting using target-city TLDs (e.g., from Meta Movement Distribution Maps) together with open spatial covariates (primarily OpenStreetMap-derived features), and a pure zero-shot fallback using the same open covariates with a national prior. Across 50 US metropolitan areas, our aggregate-calibrated gravity model recovers decay parameters and achieves downstream performance statistically equivalent to the locally-calibrated traditional Tanner baseline within a 0.01 Common Part of Commuters (CPC) margin. Furthermore, a game-theoretic Shapley analysis shows that distance decay is the dominant predictive component within the investigated gravity framework, accounting for 85.8% of the total CPC gain. This provides an interpretable, privacy-preserving pathway for zero-shot urban flow prediction without local OD surveys.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Accurate prediction of origin–destination (OD) travel flows is fundamental to transportation planning, infrastructure investment, and accessibility analysis (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Reliable OD information enables evidence-based infrastructure prioritization, travel demand forecasting, and urban mobility policy analysis across cities of all sizes. Consequently, accurate OD estimation has long been a central problem in transportation science.

Yet acquiring the localized mobility data required to calibrate OD prediction models remains a major practical challenge. Household travel surveys require substantial financial and institutional resources, limiting their coverage to a small fraction of cities worldwide (Egu and Bonnel 2020; Hussain et al. 2021). Mobile-phone and GPS trajectory records, while richer in spatial and temporal detail, are frequently unavailable because of commercial licensing restrictions and increasingly strict privacy regulations. As a result, the majority of cities—particularly those in data-scarce regions—lack reliable local OD observations, and this challenge has motivated extensive research on estimating OD flows from incomplete or alternative mobility information.

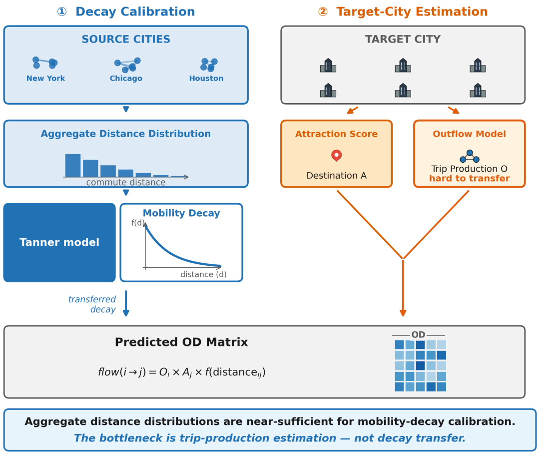

Fig. 1 Overview of the core findings: Aggregate trip-length distributions provide sufficient information for recovering city-specific distance decay $f(d)$, which acts as the dominant predictive component of gravity-based mobility prediction within the investigated framework. Prediction performance is bounded by the model's structural gravity assumptions.

A wide range of analytical and data-driven models have been developed to address this challenge. Classical spatial interaction models—including gravity models (Wilson 1971) and the radiation model (Simini et al. 2012)—are physically interpretable but require observed OD data to calibrate their distance-decay parameters. Learning-based approaches—including DeepGravity, MPGCN, and GODDAG (Simini et al. 2021; Hu et al. 2020; Rong et al. 2023a)—achieve high predictive accuracy by learning OD mappings from supervised flow matrices, imposing the same data dependency under a different computational guise. Transfer learning frameworks such as neuroGravity (Yang et al. 2026) and TransGM (Enaya et al. 2026) reduce but do not eliminate this dependency: they still require OD observations from a source city, a pretrained source-city model, or some degree of target-city structural data for adaptation. Despite their methodological diversity, these approaches all rely on some form of observed mobility information.

More precisely, existing OD prediction approaches ultimately depend on observed mobility data for model calibration, training, or transfer: gravity models require observed OD flows to calibrate distance-decay parameters; deep and graph learning methods require historical OD matrices for supervised learning; and transfer-learning approaches require mobility observations from source cities or auxiliary structural information for adaptation (Simini et al. 2021; Yang et al. 2026; Enaya et al. 2026). Although their modeling strategies differ substantially, they share the same underlying dependency on observed mobility information. This common limitation motivates the search for alternative mobility information that is both accessible and privacy-preserving.

Aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across platforms and regions, providing privacy-preserving summaries of population travel behavior. These products provide aggregated population-level trip-length distributions (TLDs) rather than individual trajectories or OD matrices. These products are publicly available, globally consistent, and designed to protect user privacy, making them substantially more accessible in data-scarce and privacy-restricted regions (Pappalardo et al. 2023). However, whether aggregate trip-length distributions contain sufficient information to replace local OD observations in the calibration of gravity models remains an open question.

This question is grounded in a fundamental mechanism: distance decay—the well-documented tendency for spatial interaction to diminish with separation distance—is a dominant determinant of travel behavior (Tanner 1961; Verma and Ukkusuri 2025). Aggregate trip-length distributions are the observable statistical outcome of these distance-decay processes at the population level. This observation motivates the following scientific hypothesis: *aggregate trip-length distributions may preserve sufficient information to recover city-specific distance-decay functions without requiring local OD observations.* This hypothesis naturally raises a fundamental question regarding the recoverability of distance-decay information from aggregate mobility summaries.

It remains unclear, however, whether aggregate trip-length distributions contain sufficient information to recover expressive, city-specific distance-decay functions under realistic urban conditions. Prior work has demonstrated that simple aggregate statistics—such as median travel time—can calibrate single-parameter spatial interaction models (Merlin 2020), and that the form of distance-decay functions substantially affects spatial interaction outcomes (Lenormand et al. 2016). Whether this extends to the two-parameter Tanner deterrence function—requiring joint estimation of power-law and exponential components while correcting for zone geometry and attraction structure—remains uncharacterized. To address this unresolved question, this study investigates the following three research questions.

RQ1 Recoverability. Can aggregate trip-length distributions recover city-specific distance-decay parameters without access to any origin-destination observations?

RQ2 Information Content. If recoverable, how much predictive information is captured by the recovered distance-decay component relative to other gravity components?

RQ3 Practical Utility. Can the recovered distance-decay information support competitive survey-free OD prediction under strict zero-shot conditions (no target-city OD observations, no trajectory data, and no target-domain adaptation)?

This paper proposes **PIGF** (Projection–Inference Gravity Framework), a survey-free framework for gravity-model calibration using aggregate trip-length distributions. The proposed framework recovers city-specific distance-decay functions from aggregate TLDs and integrates them with transferable estimates of origin production and open-data-derived destination attraction for survey-free OD prediction. Our central claim is that *aggregate trip-length distributions preserve sufficient information to recover city-specific distance-decay functions, enabling accurate gravity-model calibration and survey-free OD prediction without local OD observations.* Our empirical investigation across 50 US metropolitan areas yields the following contributions:

1. We formulate gravity calibration as an exposure-corrected inverse problem and demonstrate that aggregate trip-length distributions can recover city-specific Tanner distance-decay parameters without any OD observations, achieving high parameter recovery fidelity across diverse urban morphologies.
2. We quantify the relative importance of gravity components through factorial ablation and game-theoretic Shapley analysis, finding that the distance-decay component accounts for 85.8% of the total predictive gain within the investigated framework—an empirical finding that motivates, rather than presupposes, its central role in our pipeline.
3. We develop a survey-free calibration pipeline (PIGF) that combines aggregate mobility summaries and open spatial data to achieve downstream performance statistically equivalent to locally-calibrated baselines, providing an interpretable and privacy-preserving pathway for zero-shot urban flow prediction.

The rest of this paper is organized as follows. Section 2 reviews related work. Section **??** formalizes the problem setting. Section 4 presents the methodological design. Section 5 reports experimental results. Section 6 discusses limitations and future directions, and Section **??** addresses threats to validity.

2 Related Work

2.1 Mobility Prediction under Data Scarcity

Accurate origin–destination (OD) prediction is a fundamental task in transportation planning, accessibility analysis, infrastructure design, and urban policy because it quantifies how travel demand is distributed between locations. Traditionally, reliable OD matrices are constructed from household travel surveys, census commuting records, mobile-phone trajectories, GPS traces, or smart-card transactions. While these data sources provide detailed mobility observations, they are often expensive to collect, infrequently updated, geographically incomplete, and increasingly constrained by privacy regulations (**?**). Consequently, obtaining high-quality OD matrices remains one of the primary bottlenecks for applying mobility models, particularly in data-scarce cities and rapidly developing regions.

To alleviate this limitation, a wide range of OD prediction methods has been proposed. Classical spatial interaction models estimate mobility through analytical formulations such as gravity and radiation models (Wilson 1971; Tanner

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions.

Model Class	Decomposed Structure	Transferable Across Cities	Requires Target OD	Interpretable	Decay Explicitly Modeled
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD Prediction (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
PIGF (Ours)	Yes	Yes	No	High	Yes

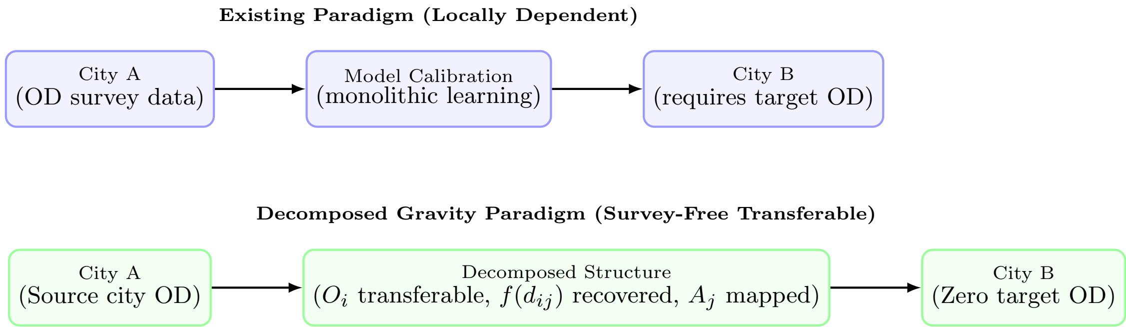

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling survey-free transfer without target-city OD surveys.

1961; Simini et al. 2012), whereas more recent approaches employ deep neural networks, graph neural networks, and transfer learning to improve predictive performance by exploiting spatial representations learned from historical mobility data (Simini et al. 2021; Hu et al. 2020; Enaya et al. 2026). These advances have substantially improved OD prediction accuracy under data-rich conditions and expanded the ability of models to capture complex spatial interactions.

Despite their methodological differences, however, most existing approaches continue to depend on some form of mobility observation during model construction or calibration. Classical gravity models require observed OD flows to estimate city-specific distance-decay parameters (Lenormand et al. 2016), while supervised deep-learning models depend on historical OD matrices or trajectory datasets for training (Simini et al. 2021). Even recent transfer-learning and zero-shot approaches typically assume that representative mobility observations are available in source cities from which transferable knowledge can be learned (Yang et al. 2026). As a result, reliable mobility observations remain an essential prerequisite for most existing OD prediction frameworks.

These limitations motivate an important question: can meaningful mobility behavior be inferred without relying on conventional OD observations? Addressing this question requires understanding not only how mobility flows are predicted, but also how the underlying behavioral mechanism governing destination choice can be identified. This issue is closely related to the estimation of distance-decay functions, which forms the focus of the next section.

2.2 Distance-Decay Modeling and Recovery

Distance decay has long been recognized as the fundamental behavioral mechanism underlying spatial interaction. Within gravity-based mobility models, it describes how interaction probability decreases with increasing travel impedance and serves as the principal constraint governing destination choice. Since the pioneering work of Tanner and Wilson, distance-decay functions have formed the behavioral core of spatial interaction models and have been widely adopted in transportation planning, accessibility analysis, migration studies, and urban mobility forecasting (Tanner 1961; Wilson 1971; ?; ?). More recent studies continue to demonstrate that the choice of deterrence function substantially influences the accuracy and interpretability of mobility prediction models (Lenormand et al. 2016; Atwal et al. 2025).

To represent heterogeneous travel behavior, numerous functional forms have been proposed, including exponential, power-law, logistic, and Tanner-type deterrence functions. Among these, the Tanner function has become one of the most widely used formulations because it simultaneously captures short-distance attraction and long-distance attenuation using only two interpretable parameters. Recent research has further extended distance-decay calibration through statistical estimation, robust optimization, sparse regression, Bayesian inference, and physics-informed modeling to improve parameter stability and transferability while preserving behavioral interpretability (?Lenormand et al. 2016; Rubio-Herrero and Muñuzuri 2023; Atwal et al. 2025).

Despite these methodological advances, existing calibration approaches share a common assumption: the availability of observed mobility flows. Classical gravity models estimate deterrence parameters directly from origin–destination matrices using maximum-likelihood estimation or least-squares optimization, while more recent data-driven approaches similarly rely on household travel surveys, mobile-phone trajectories, GPS traces, or complete OD observations to infer behavioral parameters (Wilson 1971; ?; Rubio-Herrero and Muñuzuri 2023; Yang et al. 2026). Consequently, improvements have primarily focused on how to estimate distance-decay functions more accurately, rather than whether the underlying behavioral mechanism can be recovered without mobility observations.

This limitation becomes increasingly important as privacy-preserving mobility products replace individual trajectory datasets. Although aggregate mobility summaries no longer preserve origin–destination identities, they still retain the statistical distribution of travel distances, suggesting that they may encode the latent behavioral constraints governing destination choice. Whether such aggregate information is sufficient to recover city-specific distance-decay functions, however, remains largely unexplored. Addressing this question represents a critical prerequisite for enabling survey-free gravity calibration and forms the focus of the present study.

The possibility of recovering behavioral information from aggregate mobility summaries depends fundamentally on the information preserved in emerging aggregate mobility products. We therefore next review recent developments in privacy-preserving aggregate mobility datasets and their applications in urban mobility analysis.

2.3 Aggregate Mobility Products for Privacy-Preserving Mobility Analysis

The growing availability of digital mobility data has substantially advanced the study of human movement, yet it has also intensified concerns regarding privacy protection, data governance, and the responsible sharing of individual travel records. Mobile-phone trajectories, GPS traces, and app-based location data contain detailed information about individual mobility behavior, making them highly valuable for transportation research while simultaneously raising significant privacy and ethical challenges. Consequently, recent years have witnessed a shift from publishing individual mobility records toward releasing privacy-preserving aggregate mobility products, which summarize collective travel behavior while preventing the reconstruction of individual trajectories (??).

Aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across platforms and regions, providing privacy-preserving summaries of population travel behavior. Several large-scale aggregate mobility datasets have been developed following this paradigm. Examples include Meta MDM, which provide differentially private trip-length distributions aggregated over spatial regions; Google Community Mobility Reports, which summarize temporal changes in population activity across different place categories; and aggregated mobility datasets released through the Humanitarian Data Exchange (HDX) and related humanitarian initiatives for disaster response and population monitoring. Although these products differ in spatial resolution, aggregation strategy, and intended application, they share a common objective: preserving useful mobility information while substantially reducing privacy risks associated with individual trajectory data.

Because aggregate mobility products intentionally discard origin–destination identities, previous studies have primarily employed them for descriptive mobility analysis, accessibility assessment, epidemic modeling, disaster response, and large-scale urban monitoring (**???**). Within this literature, aggregate mobility statistics are generally regarded as high-level indicators of collective movement patterns rather than as information suitable for calibrating behavioral mobility models. Consequently, their potential value for recovering the behavioral mechanisms underlying spatial interaction has received comparatively little attention.

Nevertheless, aggregate mobility products preserve an important characteristic that is directly relevant to gravity-based mobility modeling. Although they no longer identify individual origins or destinations, they retain the empirical distribution of travel distances across an urban system. Since distance-decay functions describe precisely how interaction probability varies with travel distance, these aggregate trip-length distributions may encode the latent behavioral information governing destination choice. Whether such aggregate statistics contain sufficient information to recover city-specific distance-decay functions, however, remains largely unexplored. Addressing this question is essential for enabling survey-free gravity calibration and motivates the behavioral inference framework proposed in this study.

While aggregate mobility products substantially reduce the dependence on individual trajectory data, existing mobility prediction approaches have only partially exploited the behavioral information preserved in these datasets. We therefore next review recent efforts toward survey-free and zero-shot mobility prediction and discuss their remaining limitations.

2.4 Survey-Free and Zero-Shot Mobility Prediction

The increasing scarcity of high-quality mobility observations has stimulated growing interest in survey-free and zero-shot mobility prediction. Rather than calibrating models independently for each city, these approaches aim to transfer mobility knowledge learned from data-rich regions to locations where conventional travel surveys are unavailable. By exploiting transferable spatial representations or behavioral regularities, they seek to reduce the dependence on locally collected OD observations while maintaining competitive predictive performance.

Recent studies have explored several transfer strategies. Deep learning approaches such as DeepGravity (Simini et al. 2021) demonstrate that complex nonlinear relationships between land use, population distribution, and travel demand can be learned directly from historical OD matrices, achieving substantial improvements over classical gravity models in data-rich environments. More recent graph-based and transfer-learning frameworks further improve cross-city prediction by learning transferable representations of urban mobility networks. For example, neuroGravity (Yang et al. 2026) reconstructs human mobility networks by combining neural representation learning with gravity-inspired inductive biases, while TransGM (Enaya et al. 2026) and related cross-city transfer models explicitly transfer mobility knowledge from source cities to previously unseen target regions.

These approaches significantly improve the practicality of mobility prediction under limited local data. However, they continue to rely on mobility observations during the learning process. Deep learning models require historical OD matrices or trajectory datasets for supervised training, whereas transfer-learning methods assume that representative mobility observations are available in one or more source cities from which transferable knowledge can be extracted. Consequently, although the target city may contain no OD observations, the predictive model itself remains fundamentally dependent on observed mobility behavior during model development.

An emerging line of research has begun incorporating physical principles into learning-based frameworks to improve interpretability and transferability. Physics-informed gravity models and hybrid neural architectures embed distance-decay constraints or spatial interaction mechanisms into deep learning models, reducing overfitting while preserving behavioral consistency (Atwal et al. 2025; Yang et al. 2026). Nevertheless, these methods still estimate behavioral mechanisms from observed mobility data rather than inferring them directly from privacy-preserving aggregate mobility summaries.

Therefore, despite substantial progress in survey-free and zero-shot mobility prediction, an important gap remains. Existing approaches primarily transfer behavioral knowledge learned from observed mobility data, whereas it remains unclear whether the behavioral mechanism itself can be recovered directly from aggregate mobility products without access to OD observations or trajectory records.

This remaining challenge motivates the central research gap addressed in this study. The following section synthesizes the limitations of existing literature and positions the proposed behavioral inference framework within the broader landscape of survey-free mobility prediction.

2.5 Research Gap and Positioning

The preceding review demonstrates substantial progress in origin–destination prediction, spatial interaction modeling, aggregate mobility analysis, and survey-free mobility prediction. Classical gravity models have established distance decay as the fundamental behavioral mechanism governing spatial interaction, while recent deep learning and transfer-learning approaches have significantly improved prediction accuracy by exploiting increasingly rich mobility observations. At the same time, emerging privacy-preserving aggregate mobility products have expanded access to large-scale mobility information without exposing individual travel trajectories. Collectively, these developments have substantially advanced the state of urban mobility modeling. However, they also reveal a common conceptual limitation: behavioral information is consistently treated as something that must be learned from observed mobility flows rather than inferred directly from aggregate mobility statistics.

This distinction is fundamental. Existing gravity-based approaches estimate distance-decay functions from observed origin–destination matrices through model calibration (Wilson 1971; **?**; **?**), whereas recent neural and transfer-learning methods learn transferable behavioral representations from historical mobility observations (Simini et al. 2021; Yang et al. 2026). Although these approaches differ considerably in modeling philosophy, they all assume that behavioral knowledge originates from some form of observed mobility data. Similarly, studies using privacy-preserving aggregate mobility products have primarily focused on descriptive mobility analysis, accessibility assessment, and population monitoring, rather than investigating whether these aggregate statistics themselves contain sufficient information for behavioral model calibration (**??**Meta 2026).

These observations suggest that the central challenge in survey-free mobility prediction may not be the absence of OD matrices themselves, but rather the absence of a framework capable of recovering latent behavioral mechanisms directly from aggregate mobility information. If aggregate trip-length distributions preserve the statistical signature of distance-dependent travel behavior, then behavioral model calibration may be reformulated as an inverse inference problem instead of a conventional flow reconstruction problem. Such a perspective fundamentally differs from existing survey-free approaches, which primarily transfer behavioral knowledge learned elsewhere, by investigating whether the behavioral mechanism itself can be identified without first observing origin–destination flows.

Motivated by this perspective, this study formulates survey-free gravity calibration as a behavioral inference problem. Specifically, we investigate whether privacy-preserving aggregate trip-length distributions contain sufficient information to recover city-specific Tanner distance-decay functions and whether the recovered behavioral mechanism can support competitive OD prediction when integrated with transferable production estimation and analytical destination attraction. By shifting the objective from flow reconstruction to behavioral inference, the proposed Projection–Inference Gravity Framework (PIGF) provides an interpretable and privacy-preserving alternative for mobility prediction in data-scarce environments.

The next section formalizes the problem setting and introduces the proposed behavioral inference framework.

3 Problem Formulation

Urban origin–destination (OD) prediction aims to estimate the volume of trips exchanged between spatial zones within a city. In this study, we formulate survey-free urban OD prediction as the problem of reconstructing a complete OD flow matrix using only publicly available aggregate mobility information and open spatial data, without access to any local OD observations or individual mobility trajectories.

Consider a metropolitan area partitioned into a set of N spatial zones $\mathcal{V} = \{1, \ldots, N\}$. Let $\mathbf{T} \in \mathbb{R}_+^{N \times N}$ denote the unknown OD matrix, where each element T_{ij} represents the number of trips from origin zone i to destination zone j. Let $D = [d_{ij}]$ denote the inter-zonal distance matrix, where d_{ij} is the travel distance between zones i and j. For each zone i, an open spatial feature vector x_i is available, containing publicly accessible built-environment and demographic attributes derived from sources such as OpenStreetMap and WorldPop.

Unlike conventional OD prediction settings, we assume that no target-city OD matrix, household travel survey, GPS trajectory, mobile-phone record, or other individual mobility observation is available. Instead, the only mobility information provided for the target city is an aggregate Trip-Length Distribution (TLD), denoted by $P(d)$, which summarizes the population-level distribution of travel distances without revealing any origin–destination pairs or individual trajectories. Such aggregate mobility summaries are exemplified by products such as Meta Movement Distribution Maps (MDM).

Formally, the objective is to estimate the complete OD matrix

$$\hat{T} = f(D, X, P(d)), \tag{1}$$

where $X = \{x_i\}_{i=1}^N$ denotes the set of open spatial features and $P(d)$ denotes the aggregate trip-length distribution, subject to the constraint that no target-city OD observations are used during calibration or prediction. This formulation represents a strict survey-free and zero-shot setting in which the model must infer urban mobility solely from publicly available spatial information and aggregate mobility summaries.

The principal challenge of this formulation is that the observed TLD provides only a one-dimensional statistical summary of the underlying spatial interaction process. Recovering the latent distance-dependent interaction mechanism from such aggregate observations therefore constitutes an inverse inference problem. The following section demonstrates how the separable structure of gravity models provides the theoretical basis for solving this problem.

4 Methodological Design

The methodological design of this study is grounded in a separable formulation of the classical spatial gravity framework. Under this formulation, the commuter flow between an origin zone and a destination zone is modeled as a multiplicative product of distinct spatial components. This decomposition allows us to isolate the dominant components of spatial interaction from local, city-specific variables. We structure our investigation around three core hypotheses: (1) aggregate trip-length distributions (TLDs) contain sufficient information to recover city-specific distance-decay parameters without requiring individual origin-destination records; (2) the distance-decay component is the dominant predictive component within the investigated gravity framework; and (3) integrating recovered distance-decay parameters with machine learning models for production and open-data heuristics for attraction preserves strong survey-free target-city flow prediction utility.

4.1 Gravity Model Foundation

Spatial interaction has long been modeled using gravity-based formulations (Wilson 1971; Barbosa et al. 2018), which describe travel demand as the combined effect of trip generation, destination attractiveness, and travel impedance. Owing to their physical interpretability and theoretical consistency, gravity models remain one of the most widely adopted frameworks for modeling aggregate human mobility and transportation flows.

Under the singly constrained gravity formulation, the expected flow from origin zone i to destination zone j is expressed as

$$T_{ij} = O_i \frac{A_j\, f(d_{ij})}{\sum_k A_k\, f(d_{ik})}, \tag{2}$$

where O_i denotes the total trip production generated by origin zone i, A_j represents the attraction of destination zone j, d_{ij} is the travel distance between the two zones, and $f(d_{ij})$ is the distance-decay function describing how interaction strength decreases with increasing separation distance. The normalization term ensures that the predicted outflows satisfy the production constraint, $\sum_j T_{ij} = O_i$.

A fundamental property of Eq. (2) is its **separable structure**, in which the three gravity components represent distinct mechanisms governing urban mobility. Origin production captures the total travel demand generated by each zone, destination attraction reflects the spatial distribution of opportunities, and distance decay characterizes travelers' behavioral response to travel cost. Among these components, the distance-decay function is the only term that depends explicitly on inter-zonal distance, whereas origin production and destination attraction are primarily determined by local demographic, socioeconomic, and built-environment characteristics.

This separation has an important implication for survey-free mobility modeling. Rather than reconstructing the entire OD matrix directly, gravity calibration can be reformulated as recovering only the latent behavioral component while estimating production and attraction independently from publicly available spatial information. Consequently, if aggregate mobility observations preserve sufficient information about the distance-dependent interaction mechanism, the gravity model can be calibrated without requiring locally observed OD flows.

4.2 Aggregate Trip-Length Distributions as a Projection

The gravity formulation presented in the previous section provides a natural interpretation of aggregate trip-length distributions. Rather than representing an independent mobility descriptor, a trip-length distribution can be viewed as the distance-domain projection of the underlying origin–destination interaction process. This interpretation establishes the theoretical connection between aggregate mobility observations (Meta 2026; Pappalardo et al. 2023) and the latent distance-decay mechanism that governs spatial interaction.

Let the travel distance domain be partitioned into a set of distance intervals $\{b\}_{b=1}^{K}$. The aggregate trip-length distribution is defined as

$$P(d_b) = \sum_{i,j} T_{ij}\, \mathbf{1}(d_{ij} \in b), \tag{3}$$

where $P(d_b)$ denotes the proportion of trips whose travel distance falls within interval b, and $\mathbf{1}(\cdot)$ is the indicator function identifying origin–destination pairs associated with that interval.

Unlike the OD matrix, which explicitly preserves the identities of origins and destinations, the projection removes all location-specific information and retains only the statistical distribution of travel distances. Consequently, aggregate trip-length distributions cannot directly reconstruct the complete OD matrix. Nevertheless, because the gravity formulation separates production, attraction, and distance-dependent behavioral effects, the projection continues to preserve the characteristic statistical signature of the distance-decay component.

However, the observed trip-length distribution is not equivalent to the distance-decay function itself. The projection is simultaneously influenced by urban geometry, the spatial distribution of opportunities, and the unequal number of OD pairs contributing to each distance interval. These structural effects introduce systematic **exposure biases** (Lenormand et al. 2016), implying that identical behavioral distance-decay functions may produce different aggregate trip-length distributions across cities with different spatial configurations. Therefore, directly fitting a deterrence function to the observed histogram would generally yield biased parameter estimates.

This observation transforms survey-free gravity calibration into an **inverse inference problem**. Rather than estimating the complete OD matrix, the objective becomes identifying the latent distance-dependent behavioral mechanism whose exposure-corrected projection best explains the observed aggregate trip-length distribution.

4.3 Distance-Decay Parameterization

To represent travelers' sensitivity to travel distance, we adopt the **Tanner deterrence function** (Tanner 1961), a flexible two-parameter formulation that combines power-law and exponential decay and has been shown to outperform single-family alternatives across diverse urban settings (Verma and Ukkusuri 2025). The distance-decay function is expressed as

$$f(d_{ij};\,\theta) = (d_{ij} + \delta)^{-\alpha} \exp(-\beta d_{ij}), \tag{4}$$

where $\theta = (\alpha, \beta)$ denotes the unknown behavioral parameters. The power-law exponent $\alpha \geq 0$ controls the concentration of short-to-medium distance trips, whereas the exponential parameter $\beta > 0$ governs the attenuation of long-distance interactions.

To avoid numerical singularities when travel distance approaches zero while maintaining consistent behavior across different spatial partitioning schemes, we introduce an **adaptive geometric offset**

$$\delta = 0.5\,\bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{\text{Area}_i}{\pi}} \tag{5}$$

where $\bar{R}_{\text{zone}}$ is the average equivalent radius of all spatial zones. Unlike a fixed numerical constant, the adaptive offset automatically scales with the characteristic spatial resolution of the study area, improving numerical stability while preserving the physical interpretation of the recovered distance-decay parameters. During implementation, travel distances are additionally lower-bounded by a small positive constant (10^{-6}) to prevent floating-point instability without affecting practical interzonal distances.

4.4 Exposure-Corrected Distance-Decay Recovery

Building upon the projection formulation above, distance-decay calibration is formulated as a probabilistic inverse inference problem. Instead of directly fitting the observed histogram, the proposed framework explicitly models how aggregate trip-length distributions are generated by the interaction between behavioral deterrence and spatial exposure.

For each distance interval b, let $E(b)$ denote the corresponding **spatial exposure**, defined as the total interaction opportunity associated with all origin–destination pairs whose travel distances belong to interval b. Exposure reflects purely structural characteristics of the city—including zone geometry and opportunity distribution—and is independent of travelers' behavioral preferences.

Under the gravity formulation, the expected probability that a trip falls within distance interval b is therefore

$$\pi_b(\theta) = \frac{E(b)\, f(d_b;\,\theta)}{\sum_{b'} E(b')\, f(d_{b'};\,\theta)}, \tag{6}$$

which naturally separates behavioral effects from geometric exposure.

Given an observed aggregate trip-length histogram with counts $\{y_b\}$, the observations are modeled as realizations of a multinomial distribution with probabilities defined by Eq. (6). The corresponding log-likelihood is

$$\log \mathcal{L}(\theta) = \sum_b y_b \log \pi_b(\theta), \tag{7}$$

and the city-specific distance-decay parameters are estimated by solving

$$\hat{\theta} = \arg\max_{\theta}\ \log \mathcal{L}(\theta). \tag{8}$$

To guarantee the positivity constraints $\alpha \geq 0$, $\beta > 0$, optimization is performed in the unconstrained logarithmic parameter space using the L-BFGS-B algorithm with multiple randomized initializations. The first initialization is obtained from a moment-based estimate of the observed average trip distance, while the remaining runs are generated through Gaussian perturbations to improve robustness against local optima.

Unlike conventional gravity calibration, which requires observed OD matrices, the proposed formulation estimates only the latent behavioral component. By explicitly correcting for spatial exposure, gravity calibration is transformed from a data-intensive flow-fitting problem into a statistically well-defined parameter identification problem requiring only aggregate trip-length observations. The model structure is shown in Fig. 3.

4.5 Transferable Origin Production Estimation (O_i)

Within the gravity formulation, origin production represents the total number of trips generated by each spatial zone and is conceptually independent of the behavioral interaction mechanism recovered in the previous section. While distance decay governs travelers' destination choices, origin production is primarily determined by relatively stable demographic and built-environment characteristics. This distinction enables the two components to be estimated independently.

For each spatial zone i, we construct an open spatial feature vector $\mathbf{x}_i$ consisting of publicly available variables that describe local urban characteristics. These features include population, land-use composition, point-of-interest (POI) distributions, road-network density, and other built-environment indicators derived from open geospatial datasets such as OpenStreetMap and WorldPop (Haklay and Weber 2008; Tatem 2017). Because these variables are available for virtually all cities without requiring mobility observations, they provide a transferable basis for estimating trip generation.

The mapping between spatial features and origin production is learned from source cities using a Gradient Boosting Decision Tree (GBDT) regression model (Friedman 2001),

$$\hat{O}_i = g(\mathbf{x}_i), \tag{9}$$

where $g(\cdot)$ denotes the learned nonlinear regression function and $\hat{O}_i$ is the predicted trip production for zone i. GBDT is adopted because of its strong predictive performance on heterogeneous tabular data, its ability to model nonlinear interactions without extensive feature engineering, and its robustness under cross-city prediction settings. We train the GBDT on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i).

These six features were selected from an initial 37-candidate set using a cross-city SHAP stability analysis, showing no performance degradation compared to the full feature set. This parsimonious selection is grounded in the dimensional scaling of urban physics and Occam's razor, thereby preventing the model from overfitting to source-domain geometries. Hyperparameters and details of the SHAP-based feature selection are provided in Appendix C.

Once trained on source-city data, the regression model is directly applied to the target city without further adaptation or fine-tuning. Consequently, origin production can be estimated under strict zero-shot conditions using only publicly available spatial information, eliminating the need for local travel surveys or historical OD observations.

4.6 Destination Attraction Estimation (A_j)

Unlike origin production, destination attraction reflects the relative ability of each spatial zone to attract trips from all origins within the urban system. Its value is therefore determined not only by the intrinsic characteristics of an individual zone but also by the overall spatial distribution of opportunities across the entire city. This dependence makes destination attraction considerably less transferable than origin production and limits the applicability of conventional supervised learning approaches.

Instead of learning attraction from historical destination inflows, the proposed framework estimates attraction directly from publicly available urban opportunity indicators. Let $\mathbf{x}_j$ denote the feature vector associated with destination zone j. The attraction score is estimated through a

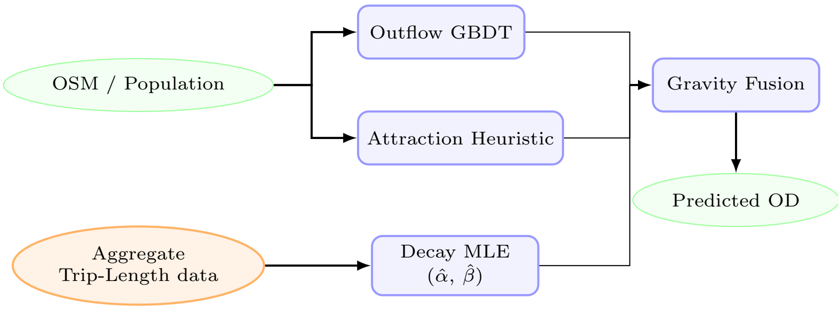

Fig. 3 Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—which provides aggregate summaries that significantly reduce privacy exposure compared with individual trajectories—feeds the Decay MLE block to recover city-specific parameters $(\hat{\alpha}, \hat{\beta})$ without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict survey-free flow matrices (T_{ij}).

training-free, scale-invariant attraction heuristic:

$$\hat{A}_j = \frac{1}{2}\left[\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right] \tag{10}$$

where $\text{minmax}(x_j) = (x_j - \min_k x_k)/(\max_k x_k - \min_k x_k + 10^{-9})$, P_j is the gridded population count, and POI_j is the OpenStreetMap POI count. This formulation normalizes absolute scale differences across cities while preserving local opportunity hierarchies, ensuring that each indicator contributes proportionally during attraction estimation.

This analytical estimation strategy offers three practical advantages. First, it is entirely survey-free, requiring only publicly available urban information. Second, it is training-free, avoiding dependence on historical destination-flow observations. Third, because the attraction scores are computed from normalized indicators, the resulting formulation remains scale-invariant, facilitating direct application across cities with substantially different spatial characteristics. In Section 5.2, the ablation analysis validates that this training-free heuristic contributes significantly to the downstream prediction accuracy.

4.7 Survey-Free Origin–Destination Flow Generation

The preceding sections independently estimate the three components required by the gravity formulation: the city-specific distance-decay function, transferable origin production, and analytical destination attraction. These components are now integrated within the gravity framework to reconstruct the complete origin–destination flow matrix.

Substituting the recovered behavioral parameters and the estimated production and attraction terms into Eq. (2) yields

$$\hat{T}_{ij} = \hat{O}_i \frac{\hat{A}_j\, f(d_{ij}; \hat{\theta})}{\sum_k \hat{A}_k\, f(d_{ik}; \hat{\theta})}, \tag{11}$$

where $\hat{O}_i$ denotes the estimated origin production, $\hat{A}_j$ represents the estimated destination attraction, and $\hat{\theta} = (\hat{\alpha}, \hat{\beta})$ are the recovered Tanner parameters.

Equation (11) preserves the production constraint $\sum_j \hat{T}_{ij} = \hat{O}_i$ while allocating trips among competing destinations according to both their relative opportunities and the recovered behavioral response to travel distance. Consequently, each gravity component contributes a distinct aspect of the final prediction. Origin production determines the total travel demand generated by each zone, destination attraction governs the distribution of opportunities across destinations, and the recovered distance-decay function captures travelers' behavioral preferences for spatial interaction.

Unlike conventional gravity calibration, which estimates these components directly from observed OD matrices, PIGF infers each component from the information source most directly associated with its underlying mechanism. Because no target-city OD matrices, household travel surveys, GPS trajectories, or mobile-phone records are required during either calibration or prediction, Eq. (11) enables survey-free and zero-shot OD generation using only publicly accessible information.

4.8 Properties of the Projection–Inference Gravity Framework

The proposed PIGF possesses several methodological characteristics that distinguish it from existing gravity calibration and supervised OD prediction approaches.

Survey-free calibration. Unlike conventional gravity models that require locally observed OD matrices for parameter calibration, PIGF estimates all gravity components without using any target-city mobility observations. Behavioral information is recovered from aggregate trip-length distributions, while production and attraction are inferred from open spatial datasets.

Component-wise inference. Instead of learning the complete OD matrix as a black-box mapping, PIGF decomposes spatial interaction into three interpretable components—origin production, destination attraction, and distance decay—and estimates each component using the information source most directly associated with its physical or behavioral meaning.

Transferability. All components of PIGF rely exclusively on publicly available or aggregate information, allowing the framework to be directly deployed in previously unseen cities without recalibration using local mobility observations. This property naturally enables zero-shot OD prediction in data-scarce environments.

4.9 Computational Complexity

The computational complexity of the proposed framework is dominated by the final gravity-based OD generation step. Distance-decay recovery operates over aggregated distance bins and therefore scales linearly with the number of distance intervals. Origin production prediction and destination attraction estimation are both linear in the number of spatial zones after model training. Consequently, the overall computational cost is dominated by evaluating pairwise gravity interactions, resulting in a complexity of

$$O(N^2),$$

where N denotes the number of spatial zones. This complexity is consistent with conventional gravity models while eliminating the computationally intensive calibration process based on observed OD matrices.

4.10 Closing Remarks

The proposed Projection–Inference Gravity Framework reformulates survey-free OD prediction as a component-wise gravity inference problem rather than a direct origin–destination prediction task. Instead of learning the entire OD matrix from historical mobility observations, PIGF independently recovers the behavioral distance-decay mechanism from aggregate trip-length distributions, estimates origin production from transferable open spatial features, and derives destination attraction analytically from publicly available urban opportunity indicators. By integrating these independently inferred components within a unified gravity formulation, the framework enables interpretable and survey-free urban OD prediction under strict zero-shot conditions.

If the proposed theoretical formulation is valid, three empirical consequences should follow. First, aggregate trip-length distributions should enable accurate recovery of city-specific distance-decay parameters. Second, the recovered behavioral component should explain a substantial proportion of OD prediction performance relative to the remaining gravity components. Third, integrating the three inferred components should produce competitive survey-free OD predictions across diverse urban environments. The following section evaluates these hypotheses through comprehensive experiments.

4.11 Data Sources

We utilize three globally available open data sources to compile features for our models: built-environment spatial descriptors (OpenStreetMap POIs and road density), gridded demography (WorldPop), and aggregate movement telemetry maps (Meta Movement Distribution Maps (Meta 2026)). Ground-truth origin-destination (OD) flows between census tracts are obtained from the LODES commuting dataset covering 50 US metropolitan areas (25 source cities for GBDT production training and 25 held-out target cities

Algorithm 1: Projection–Inference Gravity Framework (PIGF)

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate trip-length histogram $\{y_b\}_{b=1}^{K}$ (or national prior if unavailable)

Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Offline Training (Source Cities)

1. **Production Model Training**: Train the outflow model $g(\mathbf{x})$ (GBDT) on pooled source-city spatial features $\mathbf{X}^{(c)}$ and ground-truth outflows $O_i^{(c)}$.

Phase 2: Online Calibration & Zero-Shot Transfer (Target City)

2. **Decay Calibration**: Recover target-city decay parameters $(\hat{\alpha}, \hat{\beta})$ via MLE on the target city's aggregate distance-bin histogram $\{y_b\}_{b=1}^{K}$. (If unavailable, fall back to the source-derived national prior).

3. **Attraction Mapping**: Compute target city zone attractions using the training-free heuristic: $\hat{A}_j^{(t)} = \frac{1}{2}\left[\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})\right]$.

4. **Zero-Shot Target Prediction**:

   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.

   b. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.

   c. Fuse components and normalize outflows using Eq. (11) with recovered parameters $(\hat{\alpha}, \hat{\beta})$.

for testing). The target OD matrices are kept completely blind to the model components and are used exclusively as ground truth for validation.

4.12 Calibration from Aggregate Trip-Length Distributions: Meta MDM Case Study

Under the proposed theoretical framework, any aggregate data source representing the probability distribution of human movement distances is sufficient to recover city-specific distance-decay parameters. In this paper, we use the term Trip-Length Distribution (TLD) to denote aggregate travel distance distributions, including products such as the HDX Movement Distribution dataset. To evaluate this formulation under real-world telemetry constraints, we utilize Meta's Movement Distribution Maps (MDM) as a representative case study of such platform-derived aggregate mobility indicators.

Meta MDM provides aggregate travel fractions (representing empirical TLDs) at the county level grouped into 3 physical distance bins: $[0, 10$ km$]$, $(10, 100$ km$]$, and > 100 km. To harmonize these county-level travel fractions into tract geometries for calibration:

1. County-level trip-length distributions are averaged using WorldPop-derived population shares as weights: $p_{\text{metro}}(b) = \sum_{c \in \mathcal{C}} w_c \cdot p_c(b)$, where w_c is the population share of intersecting county c.
2. Tract pairs (i, j) are mapped to these 3 bins based on their centroid distances d_{ij}.
3. The predicted probability of a flow falling into Meta bin b is computed by summing and normalizing predicted tract-level flows in that bin: $P_\theta(b) = \sum_{i,j:d_{ij} \in \text{Range}(b)} \hat{T}_{ij}(\theta) / \sum_{i,j} \hat{T}_{ij}(\theta)$. This normalization bridges the continuous gravity formulation with Meta's 3-bin telemetry.

4.13 Exploratory Urban Morphology Categorization

To investigate how physical geography modulates the role of distance decay in spatial interactions, the 25 target cities are classified into four morphological groups based on their urban structure and geography (González et al. 2008). This classification is summarized in Table 2.

5 Results

5.1 Can aggregate trip-length distributions reliably recover city-specific distance-decay parameters?

We begin by evaluating direct parameter recoverability from aggregate trip-length distributions

Table 2 Morphological classification of target cities.

Morphology Group	Cities	Description
Dense / Transit ($n = 5$)	Boston, Long Beach, New York, Philadelphia, Washington DC	High-density urban cores; extensive transit networks; weaker distance decay influence.
Sprawling / Grid ($n = 10$)	Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa	Low-to-medium density; regular road grids; car-dependent mobility with uniform decay.
Polycentric ($n = 7$)	Albuquerque, Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose	Multiple employment centers; flows cluster around local nodes rather than a single center.
Coastal ($n = 3$)	Portland, Seattle, Virginia Beach	Coastline/topographic barriers constrain the road network, amplifying distance friction.

(TLDs). The recovery metrics reveal different aspects of fit quality across the two decay exponents: the power-law exponent α shows strong structural recovery across cities ($r = 0.909$), while the exponential rate β shows weaker cross-city rank correlation ($r = 0.427$) but remains close in absolute magnitude (MAE $= 0.0026$) (Table 3).

We then validate whether these recovered parameters preserve downstream predictive behavior. Under oracle outflow control, employing decay parameters recovered from aggregate TLDs yields OD prediction accuracy close to that obtained with parameters fitted on full zone-to-zone OD surveys (Table 4). Specifically, the average difference in the Common Part of Commuters (CPC) index between recovered and survey-calibrated parameters is a negligible 0.11%, with GBDT outflow prediction remaining the primary performance bottleneck in full zero-shot deployments (Table 6).

5.1.1 Validation via Trip-Length Distributions (Meta MDM Case Study)

To evaluate the practical feasibility of recovering distance decay from real-world TLDs, we apply our MLE framework to Meta's Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`) as an empirical case study across the 25 target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on LODES commuting data aggregated into 3 bins (`LODES Baseline (3-Bin)`) and 20 equal-width bins (`LODES Oracle (20-Bin)`).

As shown in Table 4, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities, compared to 0.7348 for the 3-bin survey baseline and 0.7403 for the 20-bin survey oracle.

Histogram compression is not the primary source of performance loss. When the original LODES commuting data is compressed from 20 bins to the same three physical distance bins used by Meta MDM (0–10 km, 10–100 km, > 100 km), CPC decreases only from 0.7403 to 0.7348 (-0.0055). This indicates that the coarse 3-bin representation preserves most of the information required for distance-decay recovery. In contrast, replacing the survey-derived histogram with real Meta MDM telemetry further reduces CPC from 0.7348 to 0.7080 (-0.0268). Therefore, most of the observed degradation is attributable to source mismatch and telemetry aggregation effects rather than to the limited number of bins. These results support the central claim that aggregate trip-length distributions remain highly informative for distance-decay calibration even under extremely compressed representations.

To evaluate performance under extreme data scarcity where target trip-length distributions (TLDs) are completely missing, we analyze the National Decay Fallback strategy. Under this strategy, the target city's distance-decay parameters are set to a national prior defined as the median parameter estimates fitted across the set of source cities. Under oracle outflows, this national prior model achieves a mean CPC of 0.7398, and under GBDT-predicted outflows, it yields a mean CPC of 0.6896 (as reported in Table 6).

Furthermore, we observe that the localized Meta MDM calibration is sensitive to specific local outlier dynamics. For instance, the metropolitan

Table 3 Decay parameter recovery performance metrics across the 25 target cities.

Parameter	Pearson r	MAE	Median Relative Error (%)	Cities within ±10%	Practic tion
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	Strong s across c
Exponential decay rate (β)	0.427	0.0026	0.0%	72.0%	Small despite correlat

Table 4 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies (Oracle Outflows).

Calibration Strategy	Data Source	Downstream CPC	CPC Gap vs. Oracle
Ground Truth Decay	Ground truth data	0.7411	Reference
20 equal width bins	TLD from GT data	0.7403	-0.0008
3 unequal width bins	TLD from GT data	0.7348	-0.0063
3 unequal width bins	Meta Movement Dist. Maps	0.7080	-0.0331

area of Atlanta serves as a prominent outlier in our evaluation, exhibiting a localized performance drop compared to the 3-bin LODES baseline. A detailed discussion of this outlier behavior is provided in Appendix A.2.

5.1.2 Decay Bin Sensitivity Analysis

To investigate how the granularity of distance binning affects model performance, we sweep the number of distance bins K across $\{3, 5, 10, 20\}$. The results show that predictive performance improves substantially from $K = 3$ to $K = 5$ and then exhibits diminishing returns, with mean CPC values of 0.7174 ($K = 3$), 0.7346 ($K = 5$), 0.7394 ($K = 10$), and 0.7403 ($K = 20$). Detailed comparisons and the corresponding sensitivity heatmap are provided in Appendix A.2 (Fig. A1).

5.2 To what extent does distance decay contribute to predictive performance within gravity-based mobility modeling?

To address this question on the extent of distance-decay contribution, we first evaluate the origin production component (O_i) in isolation to validate the predictive quality of our GBDT outflow models, and then implement a game-theoretic Shapley value analysis to quantify the relative contributions of each individual model component to the overall predictive accuracy.

5.2.1 Origin Production Transferability

To isolate the performance contribution of the origin production component O_i, we hold the destination attraction component A_j (computed via the OpenStreetMap heuristic) and the decay parameters (α, β) (set to the national prior) fixed. Under this controlled setting, we compare three distinct outflow prediction strategies across the 25 target cities: a naive population baseline ($O_i \propto \text{Population}_i$), the proposed GBDT-predicted outflows, and the oracle outflows derived from ground-truth data.

The empirical results indicate that the GBDT outflow prediction model (which achieves a mean CPC of 0.6896) outperforms the naive population baseline (mean CPC of 0.6758). Furthermore, this GBDT configuration achieves 93.2% of the oracle upper bound (mean CPC of 0.7398) under zero-shot transfer conditions. Additional robustness experiments analyzing production transferability are discussed in Appendix B and illustrated in Fig. A2.

Feature Selection via SHAP Stability Analysis. To construct a parsimonious model, we evaluate the predictive performance of the GBDT regressor across nested subsets of the top features identified via SHAP stability analysis. This evaluation shows that the zero-shot prediction performance plateaus when using $K = 6$ macroscopic features. The detailed feature taxonomy, the SHAP stability selection criteria, and the corresponding feature count sensitivity analyses are provided in Appendix C (specifically in Fig. B3 and Fig. B4).

5.2.2 Factorial Ablation and Shapley Value Analysis

To evaluate the relative structural importance of each gravity model component in zero-shot transfer scenarios, we perform a factorial ablation study across the 25 target cities. Each component is disabled in turn and replaced by a uniform-weight fallback, which isolates its marginal contribution to the gravity model formulation.

Based on this ablation, we execute a game-theoretic analysis using Shapley values (Shapley 1953) computed across all $2^3 = 8$ possible component coalitions. Relative to a uniform-distribution baseline (assigning equal flow weight to all zone pairs; CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2258 (accounting for 85.8% of the total gain) to the distance-decay component $f(d)$, a gain of 0.0208 (7.9% of the total gain) to the attractiveness heuristic A_j, and a gain of 0.0167 (6.4% of the total gain) to the origin production component O_i. The baseline model for the factorial ablation presented in Table 5 is the complete gravity model utilizing the national decay prior (mean CPC of 0.6896), where removing the distance-decay component $f(d)$ causes the performance to drop to a CPC of 0.4489.

5.2.3 Interpretation

Instead of learning complete OD matrices as monolithic mappings, the proposed framework explicitly isolates the behavioral mechanism governing destination choice.

The results suggest that the majority of predictive information required for gravity-based mobility prediction resides in the distance-decay component, whereas production and attraction primarily refine an already well-defined behavioral structure.

This observation provides an explanation for why aggregate trip-length distributions—despite containing no origin–destination identities—remain sufficient for recovering useful mobility information.

In other words, the success of the proposed survey-free calibration pipeline arises not because aggregate data reconstruct complete OD matrices, but because they accurately recover the dominant behavioral mechanism underlying those matrices.

5.3 Can recovering distance decay from aggregate trip-length distributions enable competitive predictive performance without access to individual OD data?

To address the final research question, we evaluate whether our proposed aggregate-calibrated gravity model can serve as a deployable, survey-free alternative to traditional locally-calibrated frameworks. We compare our model against zero-shot neural architectures and locally calibrated structures across the 25 held-out target cities. The results of these comparative evaluations are summarized in Table 6.

We deliberately exclude graph-based neural transfer models (e.g., MPGCN, GODDAG, TransGM) from this comparative evaluation, as these models inherently require either a subset of target-city OD observations for fine-tuning or full target-city mobility graphs, rendering them incompatible with the strict zero-shot, survey-free deployment setting defined in our problem formulation. Instead, DeepGravity is included as the primary neural baseline, as its design allows for pure feature-based transfer without requiring target-city OD data. Furthermore, while alternative error metrics (e.g., RMSE, MAE, KL/JS divergence) provide complementary perspectives, we report CPC as the primary evaluation metric due to its established theoretical interpretation in mobility flow reconstruction (Lenormand et al. 2016).

Our most significant empirical result is that the proposed aggregate-calibrated model achieves statistical equivalence with the locally-calibrated Traditional Tanner baseline. Specifically, under oracle outflows, a Two One-Sided Test (TOST) of equivalence confirms that our model using Meta MDM decay (CPC 0.7337) is equivalent to the locally-fitted baseline (CPC 0.7382) within a tight

Table 5 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Contribution (%)
Distance Decay $f(d)$	-34.9%	0.2258	85.8%
Destination Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

Table 6 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Survey-Free Deployment (No target OD data)				
Proposed Model (Survey-Free, Meta MDM)	0.6860	0.0391	Decomposed	6
Proposed Model (Survey-Free, National Fallback)	0.6896	0.0384	Decomposed	6
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27
Group 2: Oracle Diagnostic (True outflows $O_i^{\mathbf{true}}$)				
Proposed Model (Oracle O_i, Meta MDM)	0.7337	0.0256	Decomposed	6
Proposed Model (Oracle O_i, National Prior)	0.7398	0.0248	Decomposed	6
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27
Group 3: Locally-Calibrated Baselines (Fitted on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

margin of $\Delta_{\mathrm{eq}} = 0.01$ CPC ($p = 0.00326$), despite utilizing only aggregate telemetry. A Wilcoxon signed-rank test shows that although the minor performance difference is statistically significant (raw $p = 0.00737$), the gap is practically negligible. Additionally, the proposed model statistically outperforms the parameter-free Radiation Model baseline (Wilcoxon test with Bonferroni adjustment, $p < 10^{-6}$).

Beyond this primary equivalence result, the multi-baseline comparison yields several supporting insights. First, we observe that predicting true origin outflows (O_i) does not significantly alter the predictive accuracy of the DeepGravity model (CPC 0.7526 vs. 0.7529), as its neural architecture jointly optimizes production and destination selection. In contrast, the proposed gravity model demonstrates a strong dependency on production accuracy, where the introduction of oracle outflows narrows the performance gap, nearly matching the performance of the locally-fitted baseline.

5.3.1 Practical Implications

The proposed framework is intended for deployment in cities where conventional travel surveys are unavailable or prohibitively expensive.

Unlike existing gravity calibration approaches, PIGF requires only publicly available spatial datasets together with aggregate trip-length summaries that preserve user privacy.

Consequently, the framework provides a practical pathway for mobility estimation in data-scarce regions, privacy-sensitive environments, and rapidly developing cities where traditional OD surveys cannot be conducted routinely.

5.4 Exploratory Analysis of Urban Morphology Effects

Beyond evaluating overall predictive performance, we conduct an exploratory analysis to examine whether urban morphology may influence the effectiveness of aggregate-based distance-decay calibration. Given the limited number of target cities and the single-country scope of the dataset, these analyses are intended to generate hypotheses for future investigation rather than establish causal relationships.

Preliminary patterns suggest that the physics-constrained gravity formulation achieves its highest predictive performance in Coastal environments (mean CPC of 0.714). The relative performance gap between the proposed gravity model and the neural DeepGravity baseline narrows to 5.8% in Coastal cities, compared to a larger 10.3% gap in Dense / Transit environments. The observations are consistent with physical geographical barriers restricting alternative routes and a greater explanatory role of distance decay in commuting flows. We note, however, that these morphology-specific findings should be interpreted cautiously due to several key statistical limitations: the overall sample size of target cities is small ($N = 25$), the morphology groups are highly unbalanced (e.g., only 3 Coastal cities and 5 Dense / Transit cities), and the resulting statistical power is limited.

To supplement this categorical classification, we analyze the continuous relationship between urban density and the performance gap ($\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$). The gap correlates positively with both median employment density ($r = 0.263$) and population density ($r = 0.252$). The results are consistent with the hypothesis that dense, transit-oriented development decouples urban trips from simple Euclidean distance, allowing deep learning models to capture complex non-linear feature interactions, whereas lower-density or physically constrained environments follow more predictable distance friction behavior.

In summary, these findings should be interpreted cautiously, but they suggest that our aggregate-calibrated model provides a statistically equivalent alternative to locally-fit structures in low-density or physically constrained urban environments.

5.5 Experimental Summary

The experiments presented in this section were designed to evaluate the three research questions introduced in Section 1 through a progressive validation strategy. Rather than assessing only the final prediction accuracy, the evaluation first examined the recoverability of behavioral information from aggregate mobility summaries, then quantified its predictive importance within the gravity formulation, and finally assessed the practical utility of the complete survey-free framework under realistic deployment conditions.

For RQ1, the experiments demonstrate that aggregate trip-length distributions preserve sufficient information to recover city-specific Tanner distance-decay functions without requiring any origin–destination observations. The recovered parameters closely match conventional survey-based calibration and reproduce downstream travel behavior under both synthetic and real aggregate mobility data.

For RQ2, factorial ablation and game-theoretic attribution consistently identify the recovered distance-decay function as the dominant predictive component within the investigated gravity framework. The results indicate that most predictive information resides in behavioral distance friction rather than in production or attraction estimation, providing a mechanistic explanation for why aggregate trip-length distributions remain informative despite discarding origin–destination identities.

For RQ3, integrating the recovered distance-decay component with transferable production estimation and analytical attraction mapping enables competitive survey-free origin–destination prediction under strict zero-shot deployment. The resulting performance approaches that of locally calibrated gravity models while eliminating the need for target-city OD surveys or individual mobility trajectories.

Collectively, these experiments support a consistent scientific interpretation. The principal contribution of PIGF is not simply a new gravity calibration procedure, but a reformulation of survey-free mobility prediction as a behavioral inference problem. Instead of attempting to reconstruct complete origin–destination matrices directly, the proposed framework identifies and recovers the latent distance-dependent mechanism governing spatial interaction, then combines it with transferable production estimation and open-data-derived attraction information to reconstruct travel flows.

Table 7 Survey-Free Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.0%
Polycentric	7	0.704	0.759	7.2%
Coastal	3	0.714	0.758	5.8%

6 Discussion

6.1 Scientific Implications: Recovering Behavior Rather than Flows

The central finding of this study is not simply that PIGF achieves competitive survey-free OD prediction, but that aggregate trip-length distributions preserve sufficient information to recover the latent behavioral mechanism governing spatial interaction. Across the experimental analyses, aggregate mobility summaries consistently enabled accurate recovery of city-specific Tanner distance-decay functions, identified distance decay as the dominant predictive component within the investigated gravity formulation, and supported downstream OD prediction without requiring target-city mobility observations. Taken together, these findings suggest that behavioral information can be inferred from aggregate mobility summaries to a much greater extent than has previously been assumed.

6.1.1 From Flow Reconstruction to Behavioral Inference

Traditionally, survey-free mobility prediction has been approached as a flow reconstruction problem. Existing methods—including gravity calibration, deep learning, graph neural networks, and transfer-learning frameworks—primarily attempt to estimate complete origin–destination matrices, either by calibrating gravity parameters from observed OD flows or by learning direct mappings from historical mobility data. As a result, the availability of mobility observations remains central to most existing paradigms.

The findings of this study suggest an alternative perspective. Rather than reconstructing complete OD matrices directly, survey-free mobility prediction may instead be formulated as a behavioral inference problem. Aggregate trip-length distributions do not retain origin–destination identities, yet they preserve the statistical signature of travel-distance preferences. Recovering this latent behavioral mechanism, and combining it with independently estimated production and attraction components, provides sufficient information to reconstruct meaningful flow patterns. This interpretation shifts the emphasis from recovering individual trips to recovering the behavioral process that generates those trips.

6.1.2 Why Aggregate Mobility Remains Informative

An important implication of the experimental findings is that the usefulness of aggregate mobility data does not arise from preserving detailed spatial trajectories, but from preserving the dominant behavioral constraints governing destination choice.

The Shapley analysis shows that distance decay accounts for the majority of predictive performance within the investigated gravity framework, substantially exceeding the contributions of production and attraction estimation. This provides a mechanistic explanation for why highly compressed trip-length distributions remain informative despite discarding nearly all spatial identities. In other words, aggregate mobility summaries appear to retain the behavioral information that matters most for gravity-based interaction, even when individual travel records are no longer available. This observation also helps explain why coarse aggregate products, such as three-bin movement distributions, continue to support effective calibration despite their limited spatial resolution.

6.1.3 Implications for Physics-Informed Mobility Modeling

The proposed framework also contributes to the broader development of physics-informed mobility modeling. Recent years have seen increasing interest in combining physical principles with machine learning to improve interpretability and transferability. Most existing approaches, however, incorporate physical constraints after mobility observations have already been collected for training. In contrast, PIGF uses the physical structure of the gravity model to infer behavioral parameters directly from aggregate mobility statistics, without requiring target-city OD observations.

This distinction suggests that physical models can serve not only as predictive constraints but also as identification mechanisms, allowing latent behavioral parameters to be recovered from privacy-preserving aggregate data. More broadly, these results illustrate how interpretable physical formulations and aggregate mobility products can complement, rather than compete with, modern data-driven approaches.

6.1.4 Scientific Perspective

Taken together, the experimental evidence supports a broader conceptual interpretation of survey-free mobility prediction. Rather than viewing aggregate mobility summaries as incomplete substitutes for origin–destination matrices, they may instead be regarded as observations of the underlying behavioral processes that govern spatial interaction. Under this perspective, the central task is no longer to reconstruct every individual movement, but to identify the dominant behavioral mechanism that shapes those movements.

Within the investigated gravity formulation, this behavioral mechanism is represented by the distance-decay function. The proposed Projection–Inference Gravity Framework demonstrates that recovering this latent mechanism from aggregate mobility summaries is sufficient to enable practical survey-free gravity calibration across the evaluated metropolitan areas. Although additional validation beyond the present dataset remains necessary, the results suggest a promising direction in which behavioral inference, rather than direct flow reconstruction, becomes the primary objective of survey-free urban mobility modeling.

6.2 Methodological and Practical Implications

Beyond its scientific implications, the proposed framework also has broader methodological and practical significance for urban mobility modeling. The experimental results suggest that the proposed decomposition strategy changes not only how distance-decay parameters are estimated, but also how survey-free mobility prediction can be constructed under severe data constraints.

6.2.1 Decomposition as a Modeling Strategy

Most existing OD prediction methods optimize a single model to approximate complete origin–destination matrices. Whether based on classical calibration, deep learning, or transfer learning, these approaches generally learn a monolithic mapping between spatial inputs and mobility flows.

The proposed framework follows a different strategy by explicitly decomposing spatial interaction into production, attraction, and behavioral distance decay, allowing each component to be estimated from the information source most closely associated with its physical meaning. This decomposition offers two methodological advantages. First, it enables each component to be estimated using the most appropriate information source rather than forcing all information into a single predictive model. Second, it produces a transparent inference pipeline in which each component can be independently validated, interpreted, and improved without retraining the entire system.

6.2.2 Aggregate Data as Calibration Signals

The results also suggest a broader role for aggregate mobility products. Current aggregate datasets are frequently viewed as reduced substitutes for detailed trajectory records because they discard origin–destination identities and individual travel histories. Under this perspective, aggregate data are often considered insufficient for model calibration.

The present findings suggest a different interpretation. Aggregate mobility summaries may instead be regarded as behavioral calibration signals rather than incomplete mobility observations. Although they do not reveal where individual trips begin or end, they preserve information about the collective travel behavior governing destination choice. Consequently, aggregate mobility products can support model calibration even when they are insufficient for direct flow reconstruction. This perspective may become increasingly relevant as privacy-preserving mobility products continue to replace individual-level trajectory datasets.

6.2.3 Implications for Survey-Free Mobility Prediction

From a practical perspective, the proposed framework substantially reduces the information required for gravity-model calibration. Instead of requiring household travel surveys, GPS trajectories, or mobile-phone records, calibration relies only on publicly available spatial datasets together with aggregate trip-length distributions. These data sources are considerably easier to obtain, more consistent across regions, and inherently more compatible with modern privacy requirements.

As a result, the framework provides a feasible pathway for estimating commuting flows in metropolitan areas where conventional mobility surveys are unavailable, prohibitively expensive, or restricted by legal or privacy considerations.

6.2.4 Scope of Applicability

The proposed framework is particularly suited to deployment scenarios characterized by limited mobility observations but reasonable availability of open spatial data. Examples include rapidly growing metropolitan regions, cities with infrequent household travel surveys, and privacy-sensitive applications where individual trajectories cannot be collected or shared.

Conversely, in data-rich environments where large-scale OD observations are routinely available, high-capacity supervised neural models may still achieve higher prediction accuracy. Under such circumstances, the primary advantage of PIGF lies in its interpretability, lower data requirements, and straightforward deployment rather than maximizing predictive performance.

Taken together, these methodological and practical implications suggest that the principal value of PIGF extends beyond the proposed implementation itself. By combining interpretable model decomposition with privacy-preserving aggregate mobility information, the framework illustrates a practical pathway toward survey-free urban mobility modeling that balances predictive performance, interpretability, and realistic data availability.

6.3 Scope and Limitations

Although the experimental results consistently support the proposed behavioral inference framework, the conclusions of this study should be interpreted within the scope of the present experimental setting and modeling assumptions. Recognizing these boundaries is important for understanding both the applicability of the proposed framework and the opportunities for future research.

6.3.1 Scope of the Experimental Validation

The empirical evaluation is conducted using 50 metropolitan areas in the United States, including 25 source cities for transferable production learning and 25 held-out target cities for evaluation. These metropolitan areas encompass diverse urban morphologies, ranging from dense transit-oriented cities to sprawling, polycentric, and coastal environments. Nevertheless, they remain embedded within a common national context characterized by similar census definitions, transportation systems, and data collection procedures.

Accordingly, the present experiments demonstrate robustness across heterogeneous U.S. metropolitan environments, but do not establish that the same behavioral relationships necessarily hold under substantially different institutional, socioeconomic, or transportation contexts. Additional evaluation across countries and mobility datasets will therefore be necessary before broader conclusions can be drawn.

6.3.2 Dependence on the Gravity Formulation

The proposed framework is built upon the classical gravity formulation, in which spatial interaction is decomposed into origin production, destination attraction, and distance-dependent behavioral friction. Consequently, the interpretation of recovered behavioral information is specific to this modeling framework.

Although the experiments demonstrate that distance decay is the dominant predictive component within the investigated gravity formulation, this conclusion should not be interpreted as evidence that all urban mobility systems are governed primarily by geographical distance. In metropolitan areas where multimodal accessibility, public transportation networks, or socioeconomic interactions play a stronger role, additional behavioral mechanisms may become equally or more important. This limitation reflects the scope of the current behavioral model rather than the behavioral inference framework itself.

6.3.3 Aggregate Mobility Representation

The present study evaluates behavioral recovery using aggregate trip-length distributions represented as one-dimensional histograms. Although such representations substantially reduce privacy exposure while preserving dominant travel-distance characteristics, they inevitably discard other dimensions of human mobility, including travel direction, temporal dynamics, transport mode, and activity purpose.

Consequently, the proposed framework should be interpreted as recovering the dominant distance-dependent component of aggregate mobility rather than the complete behavioral complexity of urban travel.

6.3.4 Survey-Free Does Not Mean Data-Free

Finally, although PIGF eliminates the need for target-city origin–destination observations, it should not be interpreted as a completely data-free approach. The framework still relies on publicly available spatial information, including population distributions, built-environment characteristics, and aggregate mobility summaries, to estimate the individual components of the gravity model.

Accordingly, the contribution of this work lies in reducing the dependence on localized mobility observations, rather than eliminating data requirements altogether. The proposed framework therefore complements, rather than replaces, data-rich mobility modeling when detailed observations are available.

Taken together, these limitations define the current scope of the proposed behavioral inference framework rather than diminishing its principal findings. Within the evaluated setting, the experiments consistently demonstrate that aggregate trip-length distributions contain sufficient information to recover behaviorally meaningful distance-decay functions and support survey-free gravity calibration. Extending this framework beyond the present assumptions and evaluation domain remains an important direction for future research.

6.4 Future Research Directions

The limitations identified above also suggest several promising directions for future research. Rather than representing shortcomings specific to PIGF alone, they reflect broader challenges in survey-free mobility modeling and behavioral inference from aggregate data. Extending the present framework along these directions may further improve both the applicability and the theoretical understanding of aggregate-based mobility prediction.

6.4.1 Beyond U.S. Metropolitan Areas

The present evaluation is conducted exclusively on U.S. metropolitan areas, whose commuting behavior, transportation systems, and urban structures exhibit relatively consistent characteristics. An important next step is to evaluate whether the proposed behavioral inference framework remains applicable across cities with substantially different socioeconomic conditions, transport infrastructures, and cultural travel patterns. Cross-country validation would help determine whether the recoverability of distance-decay functions represents a universal behavioral property or depends on regional mobility characteristics.

6.4.2 Richer Behavioral Representations

The current framework models travel behavior exclusively through geographical distance. Future research may extend the behavioral inference framework to incorporate richer impedance measures, including travel time, multimodal accessibility, transport cost, and temporal congestion. Recovering multiple behavioral dimensions simultaneously may provide a more comprehensive description of urban mobility while preserving the survey-free philosophy of the proposed framework.

6.4.3 Beyond Tanner Functions

Although the Tanner formulation provides a flexible and interpretable representation of distance decay, the proposed inference framework is not fundamentally restricted to this particular functional form. Future work may investigate nonparametric behavioral recovery, neural deterrence functions, or Bayesian formulations capable of representing more complex travel preferences while maintaining physical interpretability. Such extensions would help determine whether the observed recoverability is specific to Tanner functions or reflects a broader property of aggregate mobility distributions.

6.4.4 Toward Hybrid Physics-Informed Learning

Finally, the decomposition strategy adopted by PIGF naturally suggests future integration with modern machine learning. Rather than replacing physics-based models with end-to-end neural architectures, future research may combine behavioral inference from aggregate mobility summaries with representation learning, graph neural networks, or foundation models for urban systems. Such hybrid approaches could preserve the interpretability and low data requirements of physics-informed models while benefiting from the expressive capacity of modern deep learning. This direction may ultimately bridge the current separation between interpretable gravity models and high-capacity neural mobility predictors.

Taken together, these directions indicate that the proposed framework should be viewed not as a final solution, but as an initial demonstration of behavioral inference from aggregate mobility summaries. Future developments may extend this paradigm toward richer behavioral models, broader geographical settings, and hybrid physics-informed learning frameworks, further advancing survey-free urban mobility prediction.

7 Conclusion

Urban mobility prediction has traditionally depended on localized origin–destination (OD) surveys or individual mobility trajectories, limiting the applicability of existing approaches in data-scarce and privacy-sensitive environments. Motivated by the growing availability of privacy-preserving aggregate mobility products, this study investigated whether aggregate trip-length distributions contain sufficient behavioral information to replace conventional OD observations in gravity-model calibration.

We proposed the Projection–Inference Gravity Framework (PIGF), a survey-free calibration framework that formulates gravity calibration as a behavioral inference problem. Instead of estimating complete OD matrices directly, PIGF recovers city-specific Tanner distance-decay functions from aggregate trip-length distributions through an exposure-corrected inverse inference process, integrates the recovered behavioral component with transferable origin production estimated from open spatial data and analytical destination attraction, and reconstructs commuting flows without requiring target-city OD surveys or individual mobility trajectories.

Experiments conducted across 50 U.S. metropolitan areas consistently support the three research questions posed in this study. First, aggregate trip-length distributions preserve sufficient information to recover city-specific distance-decay functions with high fidelity. Second, the recovered distance-decay component constitutes the dominant predictive mechanism within the investigated gravity formulation, accounting for the majority of predictive performance. Third, integrating the recovered behavioral component into PIGF enables competitive survey-free OD prediction under strict zero-shot deployment, approaching the performance of locally calibrated gravity models while relying only on aggregate mobility summaries and publicly available spatial information.

Beyond the proposed implementation, this work contributes a broader perspective on survey-free mobility modeling. Rather than treating aggregate mobility products as incomplete substitutes for detailed trajectories, the results suggest that they can serve as behavioral calibration signals capable of identifying the latent mechanisms governing spatial interaction. Under this perspective, survey-free OD prediction becomes fundamentally a problem of behavioral inference rather than direct flow reconstruction, providing an interpretable and privacy-preserving alternative to conventional calibration strategies.

Although the present evaluation is limited to U.S. metropolitan areas and the current framework adopts a gravity-based behavioral formulation, the findings demonstrate that meaningful urban mobility inference is possible without observing individual travel trajectories. As privacy-preserving mobility products and globally available geospatial data continue to expand, recovering latent behavioral mechanisms from aggregate observations may become an increasingly important direction for survey-free urban mobility modeling, complementing both traditional survey-based approaches and modern data-driven learning frameworks.

Appendix A Additional Experimental Results

A.1 Robustness of Aggregate-Based Distance-Decay Recovery

To evaluate the reliability of the exposure-corrected maximum likelihood estimation (MLE) workflow under non-ideal data collection conditions, we conduct a series of robustness analyses across four scenarios:

- **Bin Sensitivity Analysis**: Sweeping the number of distance bins $K \in \{3, 5, 10, 20\}$ reveals that downstream prediction performance stabilizes quickly. The mean CPC across target cities increases from 0.7174 under a coarse $K = 3$ configuration to 0.7346 for $K = 5$, 0.7394 for $K = 10$, and 0.7403 for $K = 20$ (compared to a baseline of 0.7411 computed on full, unbinned ground-truth OD matrices). This suggests that five or more distance bins provide sufficient resolution to capture city-specific decay profiles.
- **Atlanta Case Study Outlier**: Atlanta is identified as a localized outlier, where its Meta MDM 3-bin CPC under oracle outflows is 0.5761 compared to 0.7334 under the LODES 3-bin baseline. This drop is driven by the spatial interaction between the sprawling polycentric layout of the city and the coarse telemetry binning of the Meta Movement Distribution Maps (MDM) data. Excluding Atlanta from the target set increases the mean target CPC for Meta MDM from 0.7080 to 0.7135, recovering 96.4% of the corresponding LODES baseline.
- **Missing Attractiveness Data** (A_j): We simulate spatial feature scarcity by randomly removing 10%, 20%, and 50% of target-city destination attraction values A_j. The parameter recovery remains stable, yielding a mean absolute error (MAE) of $\alpha_{\mathrm{MAE}} = 0.0395$ under 20% missing data, compared to the baseline MAE of 0.0394.
- **CBD Collapse (Extreme A_j Distortion)**: To test resilience against extreme structural distortions, we replace the attraction values A_j of the top 10% most attractive zones with the city-wide mean attraction. Under this CBD collapse scenario, the estimation remains robust, yielding parameter recovery errors of $\alpha_{\mathrm{MAE}} = 0.0441$ under direct replacement and $\alpha_{\mathrm{MAE}} = 0.0486$ when the affected zone attractions are scaled down by 80%.

A.2 Robustness of Origin Production Transfer

We analyze the sensitivity of the zero-shot origin production transfer along two operational dimensions: training-set scale and prediction noise.

A.2.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot outflow predictability by retraining the GBDT model on subsets of $n_{\mathrm{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Evaluating on target cities demonstrates that zero-shot prediction performance rises steeply from $n_{\mathrm{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

A.2.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Under substantial injected noise (40%), the target CPC degrades from 0.6896 to 0.6626. While this -3.9% drop is larger than the isolated production component contribution (3.6% Shapley value), it demonstrates that even highly corrupted outflow inputs do not collapse the spatial interaction matrix, because the production-constraint normalization restricts the error propagation.

Appendix B Additional Methodological Analyses

B.1 Feature Selection Sensitivity

B.1.1 Feature Taxonomy and Candidate Set (37 dimensions)

We initially construct 37 candidate features per zone. This candidate space comprises 5 base scale

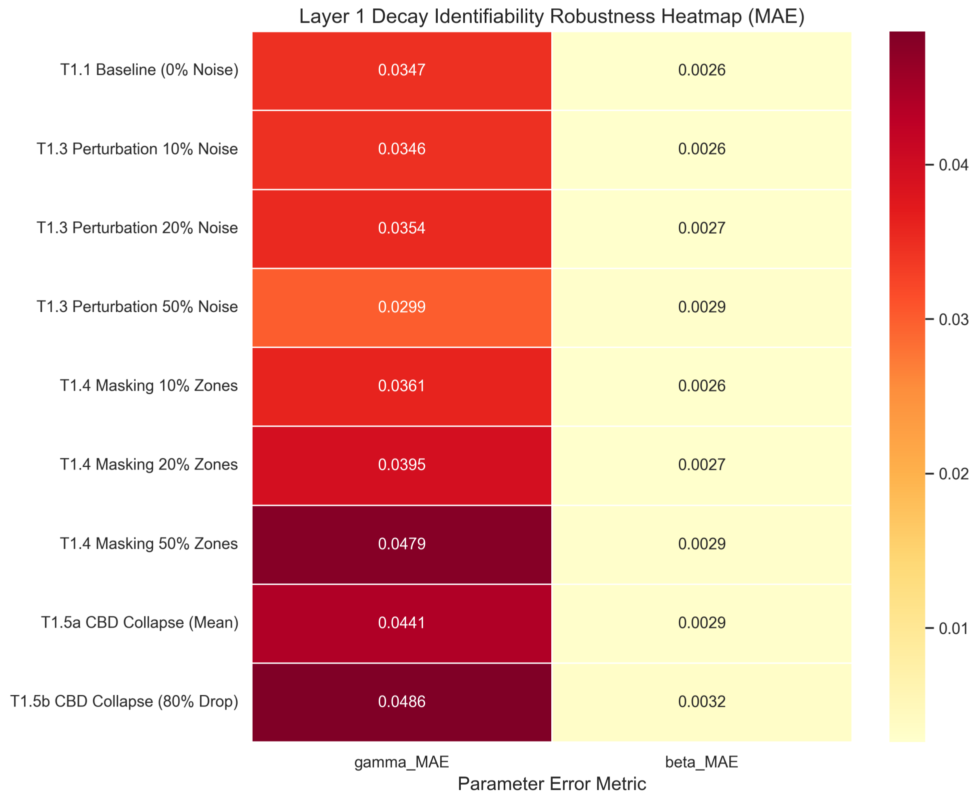

Fig. A1 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

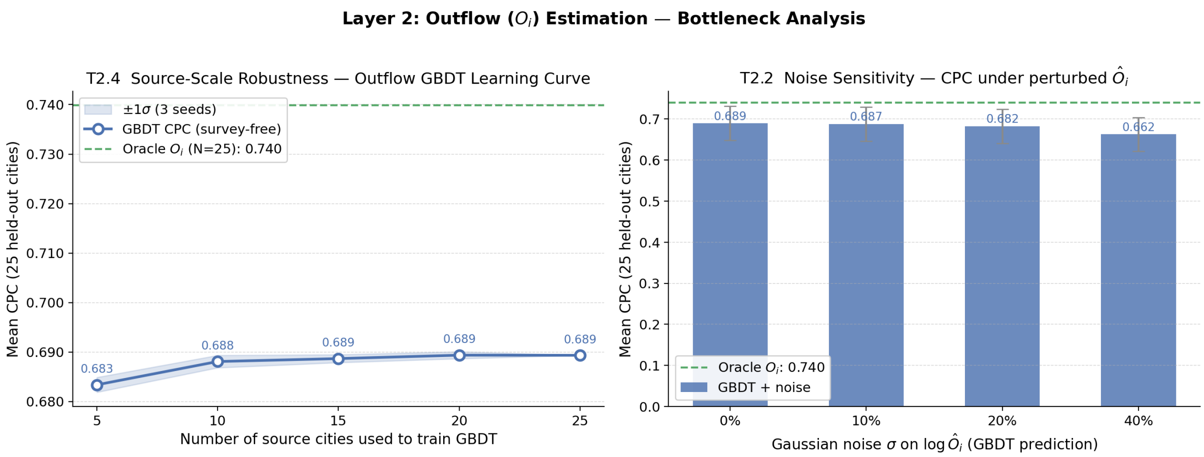

Fig. A2 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

and density variables (zone area, road density, residential population, total POI counts, and POI density) and 32 POI category count and density features (16 category counts and 16 category densities). The feature taxonomy is shown in Fig. B3.

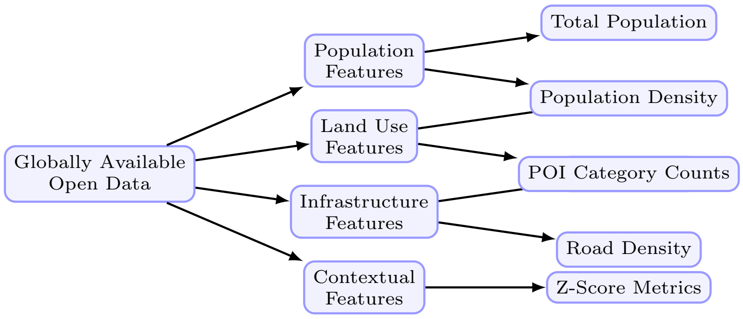

Fig. B3 Taxonomy of the globally available open data features used in the outflow model.

B.1.2 SHAP-Based Feature Stability Selection

To identify features that generalize across cities, we apply SHAP importance analysis (Lundberg and Lee 2017) to the GBDT production model. Features are ranked by a cross-city stability score $S_p = \mu_p / (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This analysis identifies 25 stable features from the 37-candidate space (32.4% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 4.5) are among the primary and most stable predictors, with no accuracy loss relative to the full candidate set.

B.1.3 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 25, 37\}$) shows that predictive performance plateaus at $K = 6$ features (CPC: 0.7080), with additional features yielding marginal utility (e.g., CPC of 0.7120 at $K = 15$). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators.

Appendix C Implementation and Reproducibility

C.1 Preprocessing and Normalization

Data preprocessing details are summarized in Table C1.

C.2 Optimization and Constraints for Distance-Decay Estimation

To solve the maximum likelihood estimation (MLE) problem for target-city decay parameters $\theta = (\alpha, \beta)$ without disaggregate flows, we adopt a robust and physically grounded initialization and optimization workflow based on closed-form moment estimates.

Good parameter initialization is critical for optimization convergence. For the exponential decay rate β, we utilize a closed-form moment estimate derived under the assumption of pure exponential deterrence and a uniform distribution of spatial opportunities. In this theoretical limit, the expected travel distance satisfies $\mathbb{E}[d] = 1/\beta$. We estimate the empirical mean trip distance $\bar{d}_{\text{obs}}$ from the observed distance-bin histogram:

$$\bar{d}_{\text{obs}} = \sum_{k=1}^{K} b_k \cdot m_k \tag{C1}$$

where b_k is the observed proportion of trips in bin k, and m_k is the midpoint of distance bin k (for the open-ended final bin, we use 1.5 times the lower boundary as a surrogate midpoint). The

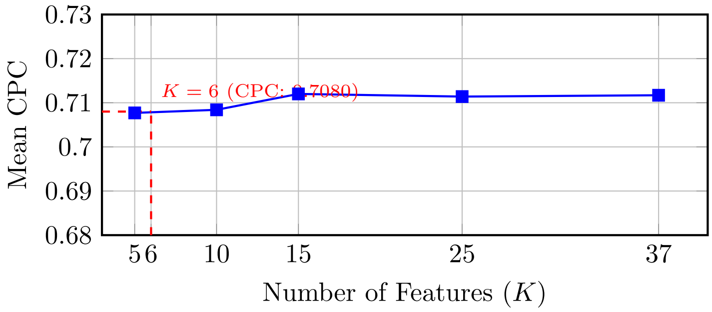

Fig. B4 SHAP feature count sweep over the 37-candidate feature set. While utilizing more than 6 features yields marginal accuracy gains, a parsimonious selection of 6 features is preferred to maintain generalization across diverse target-city environments.

Table C1 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	**Treatment**	**Rationale**
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

initial value $\beta^{(0)}$ is set as:

$$\beta^{(0)} = \frac{1}{\bar{d}_{\text{obs}}} \tag{C2}$$

For the power-law exponent α, the initial value is set statically to $\alpha^{(0)} = 1.0$, which corresponds to the center of the empirically observed range $[0.5, 2.0]$ in classical spatial interaction modeling literature.

The positivity constraints $\beta > 0$ and $\alpha \geq 0$ are strictly enforced during optimization via a parameter transformation. We define log-transformed parameters:

$$\tilde{\beta} = \log \beta, \quad \tilde{\alpha} = \log \alpha \tag{C3}$$

and optimize the objective in the unconstrained log-transformed space $\theta_{\log} = (\tilde{\alpha}, \tilde{\beta}) \in \mathbb{R}^2$.

To guard against convergence to local optima due to the potential non-convexity of the exposure-corrected likelihood surface (especially under highly binned or noisy data), we run the L-BFGS-B optimization algorithm from 10 random restarts. The first run is initialized at the theoretical values $\theta_{\log}^{(0)} = (\log \alpha^{(0)}, \log \beta^{(0)})$. The remaining 9 runs are initialized by perturbing this central starting point with Gaussian noise:

$$\begin{aligned} \tilde{\alpha}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \alpha^{(0)}, 0.25), \\ \tilde{\beta}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \beta^{(0)}, 0.25) \end{aligned} \tag{C4}$$

The optimization is executed for each restart run, and the parameter set yielding the highest likelihood value $\mathcal{L}(\theta)$ is recovered via exponential back-transformation: $\hat{\alpha} = \exp(\tilde{\alpha}^*)$ and $\hat{\beta} = \exp(\tilde{\beta}^*)$.

C.3 Hyperparameter Settings

The GBDT hyperparameter configurations are detailed in Table C2.

Appendix D Mathematical Details

Detailed derivations of the exposure-corrected projection and adaptive scale offsets are provided in the primary text (Section 4).

Table C2 GBDT Model Hyperparameter Configurations

Hyperparameter	**Outflow Model (O_i)**	**Attraction Model (A_j, ablation only)**
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	2	2
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Appendix E List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table E3.

References

Atwal KS, Anderson T, Pfoser D, et al (2025) Commuting flow prediction using OpenStreetMap data. Computational Urban Science 5(1):2. https://doi.org/10.1007/s43762-025-00161-5

Barbosa H, Barthelemy M, Ghoshal G, et al (2018) Human mobility: Models and applications. Physics Reports 734:1–74. https://doi.org/10.1016/j.physrep.2018.01.001

Egu O, Bonnel P (2020) How comparable are origin–destination matrices estimated from automatic fare collection, origin–destination surveys and household travel survey? an empirical investigation in lyon. Transportation Research Part A: Policy and Practice 138:267–282. https://doi.org/10.1016/j.tra.2020.05.021

Enaya A, Zhong C, Batty M, et al (2026) TransGM: Transferable gravity models for cross-city policy transfer. Computers, Environment and Urban Systems https://doi.org/10.1016/j.compenvurbsys.2026.102455

Friedman JH (2001) Greedy function approximation: a gradient boosting machine. Annals of statistics pp 1189–1232

González MC, Hidalgo CA, Barabási AL (2008) Understanding individual human mobility patterns. Nature 453(7196):779–782. https://doi.org/10.1038/nature06958

Haklay M, Weber P (2008) Openstreetmap: User-generated street maps. IEEE Pervasive Computing 7(4):12–18

Hu J, Yang G, Li K, et al (2020) Predicting origin-destination flow via multi-perspective graph convolutional network. In: IEEE 36th International Conference on Data Engineering (ICDE), pp 1818–1821, https://doi.org/10.1109/ICDE48307.2020.00178

Hussain E, Bhaskar A, Chung E (2021) Transit od matrix estimation using smartcard data: Recent developments and future research challenges. Transportation Research Part C: Emerging Technologies 125:103044. https://doi.org/10.1016/j.trc.2021.103044

Lenormand M, Bassolas A, Ramasco J (2016) Systematic comparison of trip distribution laws and models. Journal of Transport Geography 51:158–169. https://doi.org/10.1016/j.jtrangeo.2016.02.003

Lundberg SM, Lee SI (2017) A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30:4765–4774. https://doi.org/10.48550/arXiv.1705.07874

Merlin LA (2020) A new method using medians to calibrate single-parameter spatial interaction models. Journal of Transport and Land Use 13(1):49–70. https://doi.org/10.5198/jtlu.2020.1614

Table E3 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations
PIGF	Projection–Inference Gravity Framework		

Meta (2026) Movement distribution maps. https://ai.meta.com/ai-for-good/datasets/movement-distribution-maps/, aI for Good Datasets. Accessed: 27 June 2026. No DOI assigned.

Pappalardo L, Manley E, Sekara V, et al (2023) Future directions in human mobility science. Nature Computational Science 3:588–600. https://doi.org/10.1038/s43588-023-00469-4

Rong C, Feng J, Ding J, et al (2023a) GODDAG: Generating origin-destination flow for new cities via domain adversarial training. IEEE Transactions on Knowledge and Data Engineering 35(10):10048–10057. https://doi.org/10.1109/TKDE.2022.3218526

Rong C, Wang H, Li Y (2023b) Origin-destination network generation via gravity-guided gan. URL https://arxiv.org/abs/2306.03390, arXiv preprint; no DOI assigned, arXiv:2306.03390

Rubio-Herrero J, Muñuzuri J (2023) Sparse regression for data-driven deterrence functions in gravity models. Annals of Operations Research 323(1–2):153–174. https://doi.org/10.1007/s10479-022-04803-y

Shapley LS (1953) A value for n-person games. In: Contributions to the Theory of Games, vol 2. Princeton University Press, p 307–317, classic book chapter; no DOI assigned

Simini F, Gonzalez M, Maritan A, et al (2012) A universal model for mobility and migration patterns. Nature 484:96–100. https://doi.org/10.1038/nature10856

Simini F, Barlacchi G, Luca M, et al (2021) A deep gravity model for mobility flows generation. Nature Communications 12(1):5647. https://doi.org/10.1038/s41467-021-26752-6

Song C, Qu Z, Blumm N, et al (2010) Limits of predictability in human mobility. Science 327(5968):1018–1021. https://doi.org/10.1126/science.1177170

Tanner J (1961) Factors affecting the amount of travel. Road Research Technical Paper 51, Road Research Laboratory, London

Tatem AJ (2017) WorldPop, open data for spatial demography. Scientific Data 4:170004. https://doi.org/10.1038/sdata.2017.4, worldPop gridded population datasets available at https://www.worldpop.org/

Verma R, Ukkusuri SV (2025) What determines travel time and distance decay in spatial interaction and accessibility? Journal of Transport Geography 122:104061. https://doi.org/10.1016/j.jtrangeo.2024.104061

Wilson A (1971) A family of spatial interaction models, and associated developments. Environment and Planning A 3(1):1–32. https://doi.org/10.1068/a030001

Yang J, Huang S, Huang Z, et al (2026) Transferable human mobility network reconstruction with neurogravity. Nature Computational Science 6:630–641. https://doi.org/10.1038/s43588-026-01003-y